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Unit - II

Smart Data Querying with SQL & PL / SQL.

Syllabus :

SQL : Data types, DDL, DML, DCL and TCL commands (Create User, Grant Revoke Queries)

Tables : Create, Insert, Update, Delete, Alter, Drop, SELECT, Joins, Views (Create, Update, Drop); Aggregation functions, set operations, SQL Operators, SQL Predicates, Nested queries.

PL/SQL : Stored ~~prod~~ procedures, functions, triggers, assertions, roles, and privileges.

* SQL - Structured Query Language

- It is a Standard language. Used to retrieve, and manage data in databases.
- SQL is especially used with relational database, where data is organized in tables (rows & columns)

* Data Types

- INT / Integer - whole numbers.
  e.g age INT
- VARCHAR(n) - text of variable length
  e.g name VARCHAR(50)
- CHAR(n) - fixed-length text
- DATE - date values
- FLOAT / DECIMAL - decimal numbers.
- BOOLEAN - true / false.

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* DDL (Data Definition Language)

DDL commands are used to define or change the structure of database objects.

- DDL commands.

1. CREATE - Create database objects.

Syntax:

```
CREATE TABLE Student (
    id INT,
    name VARCHAR(50)
);
```

2. ALERT - modify table structure.

Syntax:

```
ALERT TABLE Student ADD age INT;
```

3. DROP - delete database objects.

Syntax:

```
DROP TABLE Student;
```

4. TRUNCATE - remove all records but keep table structure

Syntax:

```
TRUNCATE TABLE Student;
```

DML (Data Manipulation Language)

- DML commands are used to manage data inside tables.

- DML commands

1. INSERT - add new records

Syntax :

```
INSERT INTO student VALUES (1, 'Ravi', 20);
```

2. UPDATE - modify existing data

Syntax :

```
UPDATE student SET age = 21 WHERE id = 1;
```

3. DELETE - remove records.

Syntax :

```
DELETE FROM student WHERE id = 1;
```

* DCL (Data Control Language)

- DCL controls access and permissions.
- primarily for security purpose

1. GRANT - give permissions

Syntax:

GRANT SELECT ON student TO user1;

2. REVOKE - remove permissions.

Syntax:

REVOKE SELECT ON student FROM user1;

* TCL (Transaction Control Language)

- TCL manages transactions (a group of SQL statements)
- They are essential for maintaining the ACID properties (Atomicity, Consistency, Isolation, Durability) of a database.
- TCL Commands.

1. COMMIT - Save changes permanently.

Syntax:

COMMIT;

2. ROLLBACK - undo changes

Syntax:

ROLLBACK;

3. SAVEPOINT - Set a point to rollback to

Syntax: SAVEPOINT savepoint_name;

Example :

```
SAVEPOINT S1;
ROLLBACK TO S1;
```

TABLES

- In Sql, a table is a store data in rows and columns.

• Each row = record

Each column =

Example :

```
CREATE TABLE Employee (
    emp_id INT,
    emp_name VARCHAR(50),
    Salary DECIMAL(10,2)
    );
```

- Operations on the Table.

1.	Create	7.	SELECT
2.	Insert	8.	Joins
3.	Update	9	views
4.	Delete		
5.	Alter		
6.	Drop		

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* **SELECT** - Used to retrieve data from table.

example :

```
SELECT * FROM Employee;
SELECT emp-name, salary FROM
Employee WHERE salary > 30000;
```

* **Views** - A view is a virtual table created from a query.

example :

- Create view :

```
CREATE VIEW Highsalary AS
SELECT emp-name, salary
FROM Employee
WHERE salary > 50000;
```

- Drop / update - view

```
DROP VIEW Highsalary;
```

* **Joins** - Joins combine data from multiple tables.

- **Types of Joins**

1. INNER Join - matching records only

2. LEFT JOIN - all from left table
3. RIGHT JOIN - all from right table
4. FULL JOIN - all records.

example:

```
SELECT e.emp_name, d.dept_name
FROM Employee e
INNER JOIN Department d
ON e.dept_id = d.dept_id;
```

* Aggregation Functions.

- An Aggregation function in SQL perform a calculation on a set of values (multiple rows) and return a single value as a result.

. Common Aggregate Functions.

1. AVG(): Return the average value of a numeric column.

2. COUNT(): Counts the number of rows or non NULL values in a column.

3. MAX(): Returns the largest (maximum) value in a column.

4. MIN(): Returns the smallest (minimum) value in a column.

5. SUM(): Return the total sum of the values in a numeric column.

- Example :

```
SELECT AVG (salary) FROM Employee;
```

* Set Operations.

- It combines the results of two or more SELECT statement into a single result set.

- Types of set operations.

1. UNION : combines results (no duplicates)
2. UNION ALL : Includes duplicates
3. INTERSECT : Common records
4. EXCEPT / MINUS : difference

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example:

```
SELECT name FROM A
UNION
SELECT name FROM B;
```

* SQL Operators

- Operators perform operations in conditions

- Types:

1. Arithmetic operators
   + : Addition
   - : Substraction
   * : Multiplication
   / : Division
   % : Modulus

2. Comparison Operator
   = : Equal to
   > : Greater than
   < : Less than
   >= : Greater than or equal to
   <= : less than or equal to
   != : Not equal to

3. Logical Operator :
AND
OR
NOT

4. Special Operator :
LIKE
IN
BETWEEN
IS NULL

* SQL Predicates

- A predicate is a condition or expression that evaluates to Boolean value (either True, False or Unknown)
- Predicates are primarily used in WHERE, HAVING, and ON clauses of SQL statement to filter data, determining which rows are included in the result set.

example

```
WHERE age BETWEEN 18 AND 25
WHERE name LIKE 'A%'
WHERE dept_id IN (10,20)
```

* Nested Queries (subqueries)

- A query inside another query.
- SQL query embedded within another SQL Statement (the outer query)

• Key Concepts

1. Enclosed in Parentheses :
Subqueries must always be enclosed within parantheses ().

2. Location :
They can be placed in various clauses of the outer query, including - SELECT, FROM, WHERE and HAVING.

• Example

```
SELECT emp_name
FROM Employee
WHERE Salary >
(SELECT AVG(salary) FROM Employee);
```

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* PL/SQL :

- PL/SQL (Procedural Language/SQL) is an extension of SQL used for procedural programming.

- Features :-

1. Variables
2. Loops
3. Conditions
4. Exception handling

- Syntax :-

```
DECLARE
    ---- variable declaration
BEGIN
    ----- executable statements
EXCEPTION
    ---- error handling
END;
/
```

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Stored procedures

- A Stored procedure is a named PL/SQL program Stored permanently in the database that is used to perform a specific task.

Why Stored Procedure used ?

1. Reusability - write once, use many times
2. Better performance - executed on the database server
3. Security - Users can execute procedures without direct table access.
4. Reduced network traffic - multiple SQL Statements run as one block.

Syntax :

```
CREATE OR REPLACE PROCEDURE procedure_name
(
  parameter_name datatype [IN | OUT | INOU
)
AS
BEGIN
    ---- SQL/PL-SQL Statements
END;
/
```

* Procedure with Parameter (PL/SQL)

A procedure with parameter is a stored procedure that accepts values from the user and / or return values while executing its logic.

- Three Types of Parameters :

1. IN
2. OUT
3. IN OUT

1. IN Parameter :- Used to pass input values to the procedure.

2. OUT Parameter :- Used to return a value from the procedure.

3. IN OUT Parameter :- Used to accept input and return modified output.

* Functions

- A Functions returns a value using RETURN Statements.
- Functions are mainly used for calculations and computation.

- Syntax :

```
CREATE OR REPLACE FUNCTION function_name
(
  parameter_name datatypes
)
RETURN return_datatypes
As
BEGIN
    ---- Statements.
  RETURN value ;
END ;
/
```

- Example

```
CREATE OR REPLACE FUNCTION get_bonus
(
  Salary NUMBER
)
RETURN NUMBER
AS
BEGIN
    RETURN Salary * 0.10 ;
END ;
/
```

* Triggers

- Triggers execute automatically when an event occurs. (INSERT, UPDATE, DELETE)

- Ty They can be defined:
  - BEFORE the event
  - AFTER the event

* Assertions

- Assertions are conditions that must always be true for the database

e.g Salary > 0
(conceptual topic, rarely implemented directly in SQL)

* Roles

Roles are collections of privileges.

example :

```
CREATE ROLE manager;
GRANT SELECT, INSERT ON EMPLOYEE
  TO manager;
```

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* Privileges.

Privileges define what actions a user can perform.

Types:

1. System privileges (CREATE TABLE)

2. Object privileges (SELECT, INSERT)